

Introduction

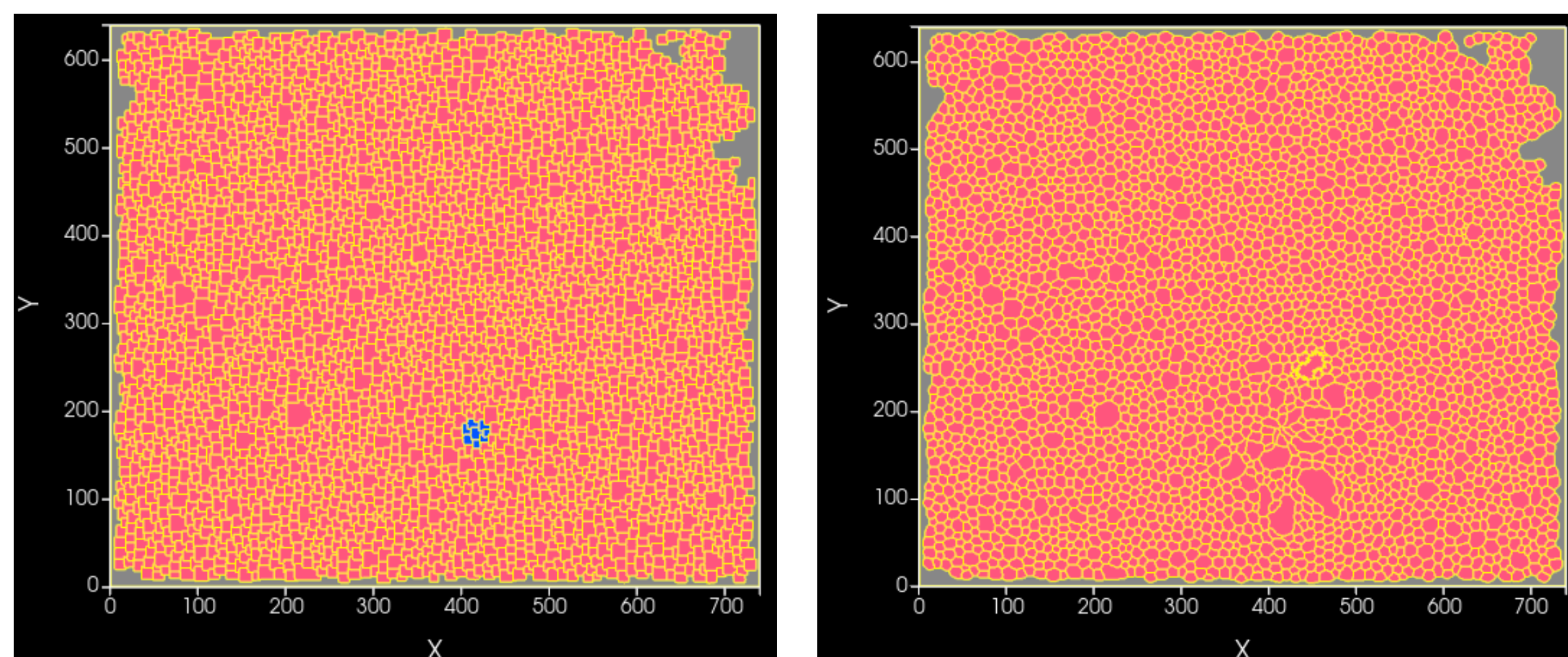
- Age-related Macular Degeneration (AMD) is the leading **cause of vision loss** in the aging population.
- There is **no effective treatment** currently available for AMD.
- We hypothesize that **aging and disease** processes impose **stresses on the Retinal Pigment Epithelium (RPE)**, altering its morphology and contributing to AMD development.

Purpose

- Understanding of RPE repair mechanisms
- Informing future therapeutic strategies

CompuCell3D (CC3D)

- CompuCell3D (CC3D) is a Python-based simulation tool used to build, test, run and visualize multiscale and multi-cell simulations.



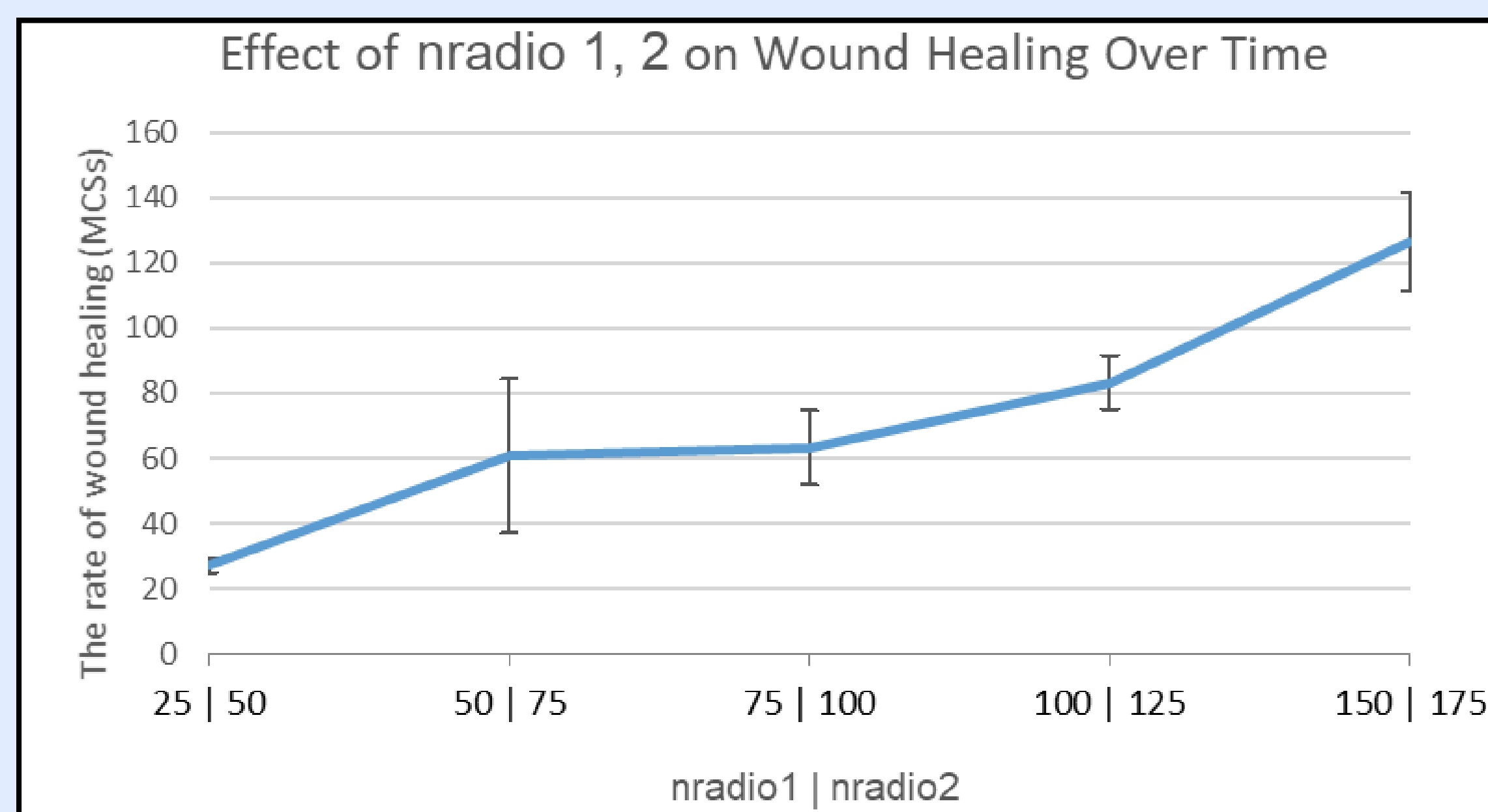
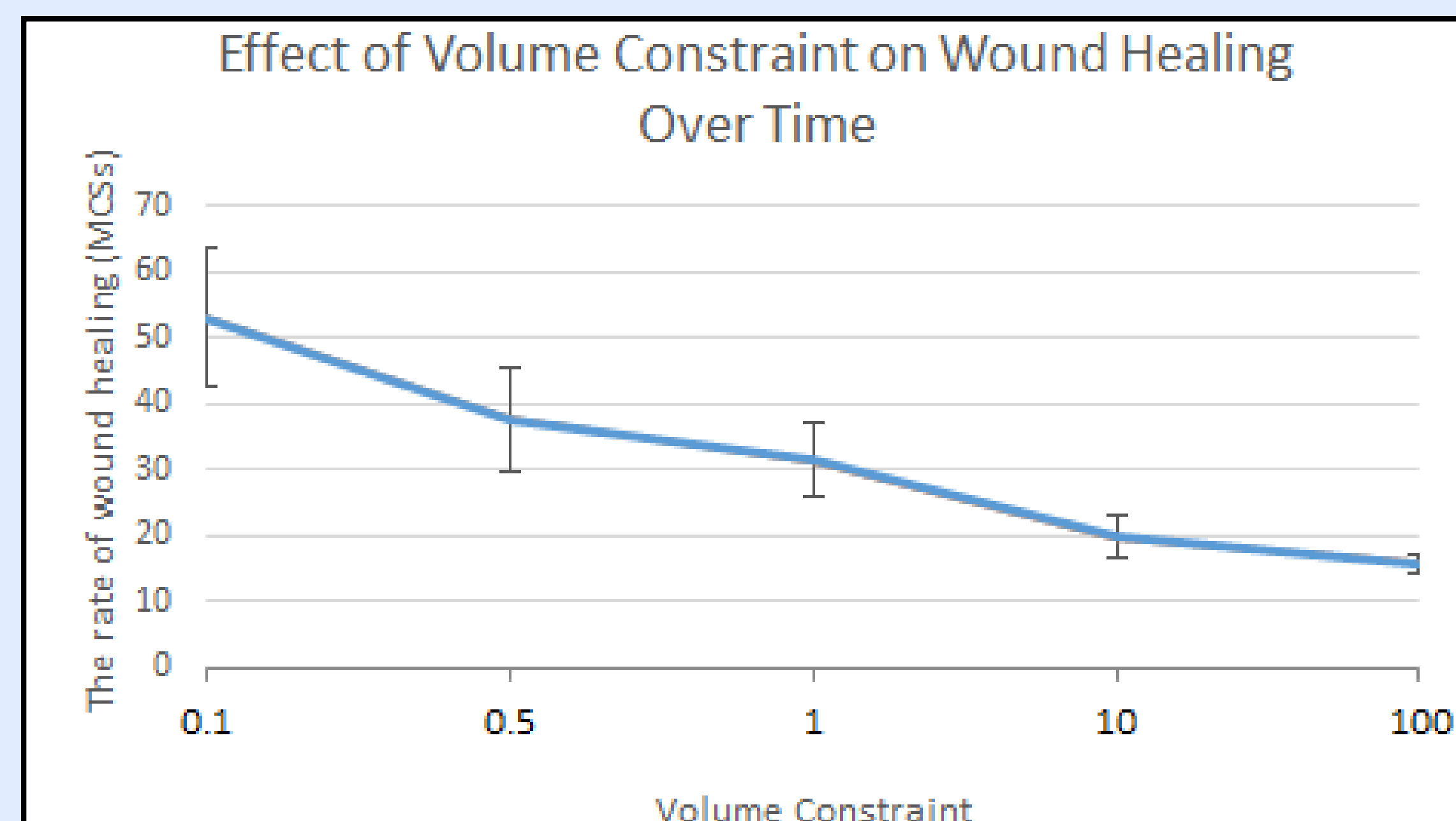
The simulations of wound healing process captured from CC3D

Fig.1: The wound appears. Fig.2: The wound is fully healed

Methods

- **Quantitatively describing the relationship** between RPE morphology, age, and disease status in AMD progression → **Developing a mathematical model** for RPE morphogenesis → **Utilizing CC3D to model** the RPE wound healing process.
- We adjusted 2 parameters to investigate its effects on the wound healing process: **volume constraint** and **nratio 1 and nradio 2**
- In CC3D, for each parameter, we conducted five simulation runs, each with a different random number seed. Within each run, we performed five rounds of wound healing simulations. The healing rate was calculated as the **mean ± standard deviation** of these 25 wound-healing rounds.

Results



Discussion

- Higher volume constraints result in a faster and more stable healing process, as evidenced by the reduced number of steps required for healing and the lower standard deviation in the results.
- **radio1** directly affects the wound radius, with higher values leading to larger wounds, and **nratio2** impacts the wound's affected area, with increased values resulting in more cells interacting with and closing the wound.

Conclusion

- This result is the first step in understanding the wound healing process. We can confirm that changing the volume constraint significantly affects the healing process.
- These insights could be instrumental in developing better therapeutic strategies for conditions like AMD where effective wound healing in the retinal pigment epithelium is crucial.

Future Direction

- We will explore the effects of varying additional parameters **cell-cell adhesion** and **cell-wound interaction**, which influence the overall cohesion and response of cells or between cells and wounds during the healing process.

Acknowledgement

- Special thanks to Dr. Yi Jiang for her mentorship, and Ms. Karina Mazzitello and Mr. Joel Vanin for their support. We also appreciate the Math Path Program for their valuable resources. We are particularly grateful to Alfred P. Sloan Foundation for their generous grant (G-2021-16960)